A DB-LINEAR PROGRAMMABLE GAIN AMPLIFIER (PGA) WITH MIXED-SIGNAL CONTROL

INTRODUCTION

- The Need for VGAs: Variable Gain Amplifiers (VGAs) are critical in wireless receivers to maintain constant output power despite varying input signal levels.
- Design Challenges: Maximizing dynamic range while minimizing power consumption is a primary trade-off.
- Achieve a wide gain tuning range with precise dB-linear characteristics.
- Target low-power IoT receiver application
- Proposed Solution: A robust, cascaded unit cell-based PGA using MOS transistors in the sub-threshold region.

THE EVOLUTION OF GAIN CONTROL

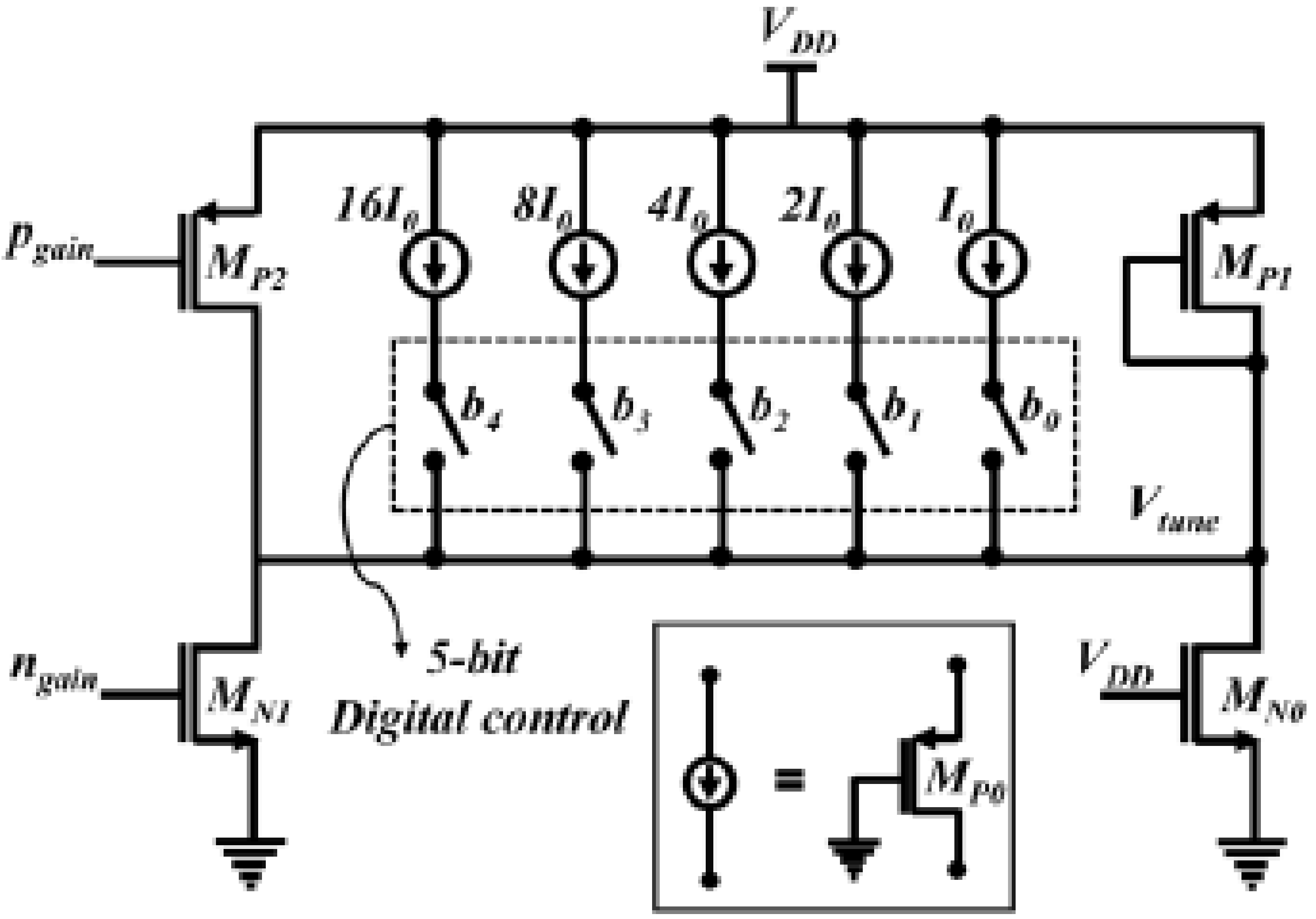
- **The Necessity of Variable Gain**: In wireless communications, input signals vary wildly in strength (Dynamic Range) depending on the distance from the transmitter.
- **The Role of AGC**: Automatic Gain Control (AGC) loops are essential in receivers to maintain a constant output power.

Historical Shift:

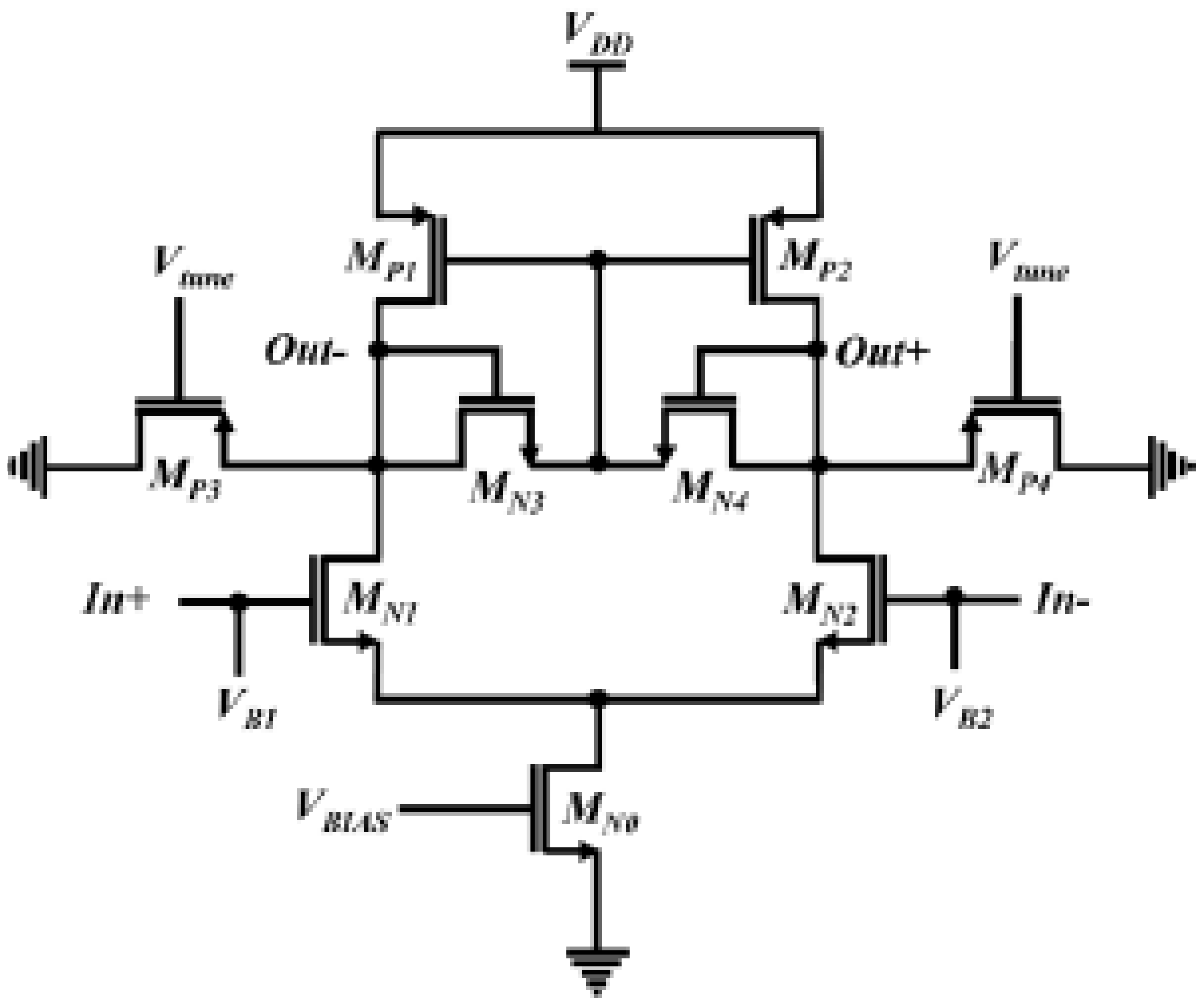
- Best of Both Worlds: Unlike most chips that are either just digital or just analog, this one allows control via both digital computer codes and analog voltage, making it flexible for many different devices.

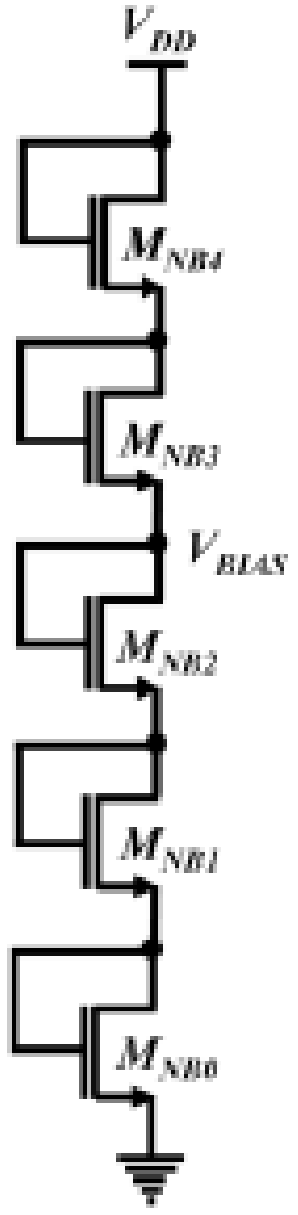
- Huge "Volume" Range: It can handle a massive range of signal strengths (76 dB range)—amplifying very weak signals or handling strong ones without making errors

Gain tuning Circuit



PGA unit cell and Bias circuit





Pgain= 1.1V

ngain=0V

Vbw=800mv

Vin_dc=562mv

Vtune=150mv−450mv

V_bias=590mv

DCOC and Buffer circuit

Pgain= 1.1V

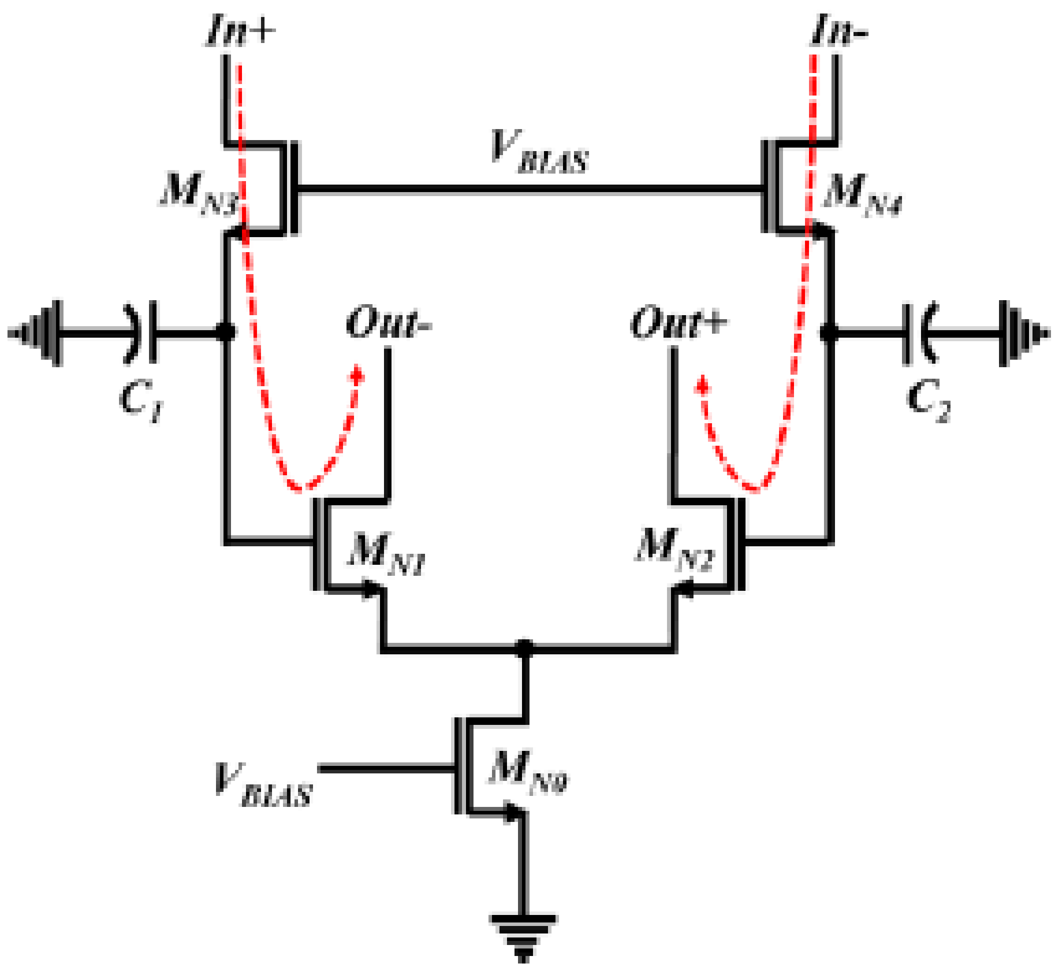
ngain=0V

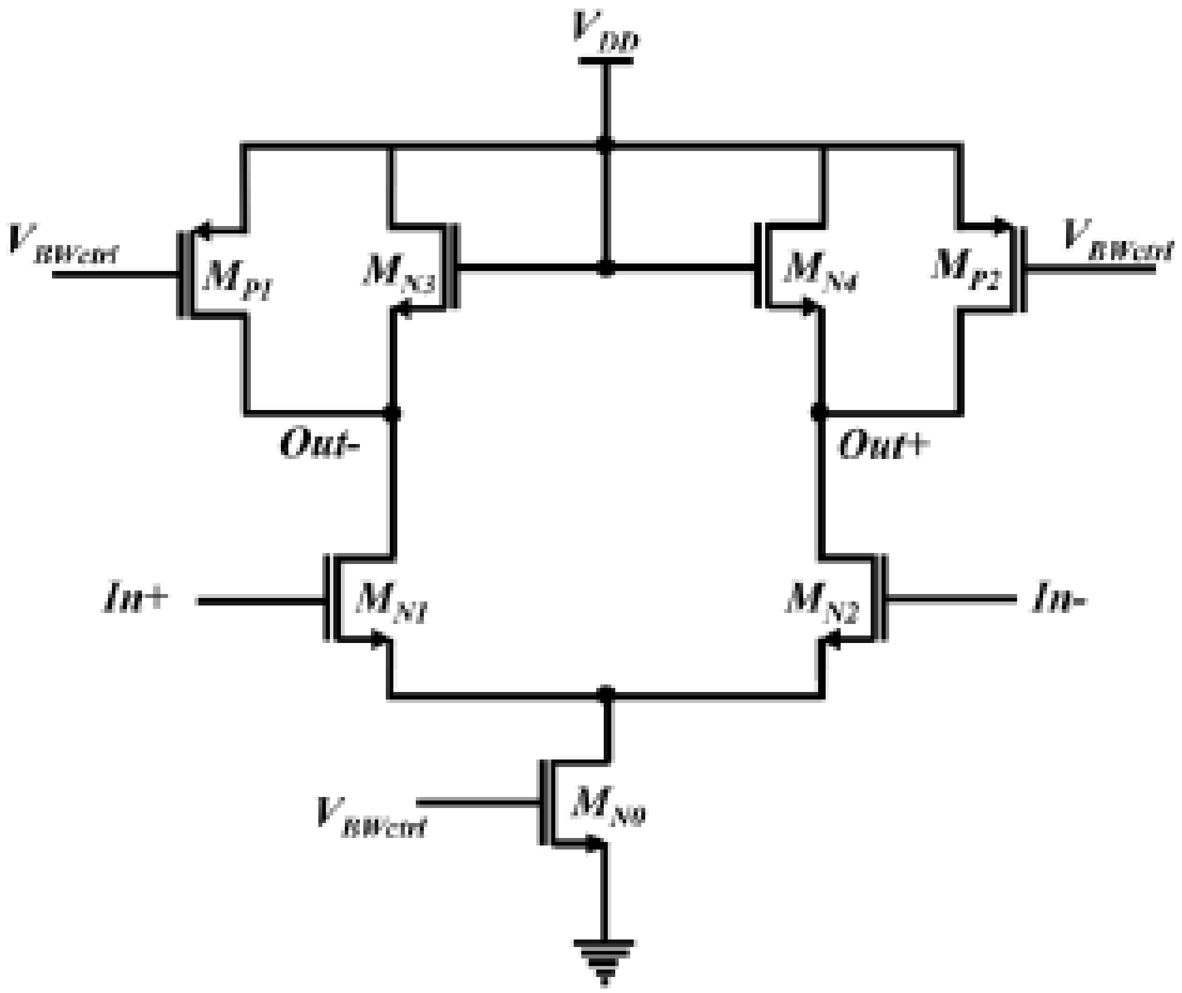
Vbw=800mv

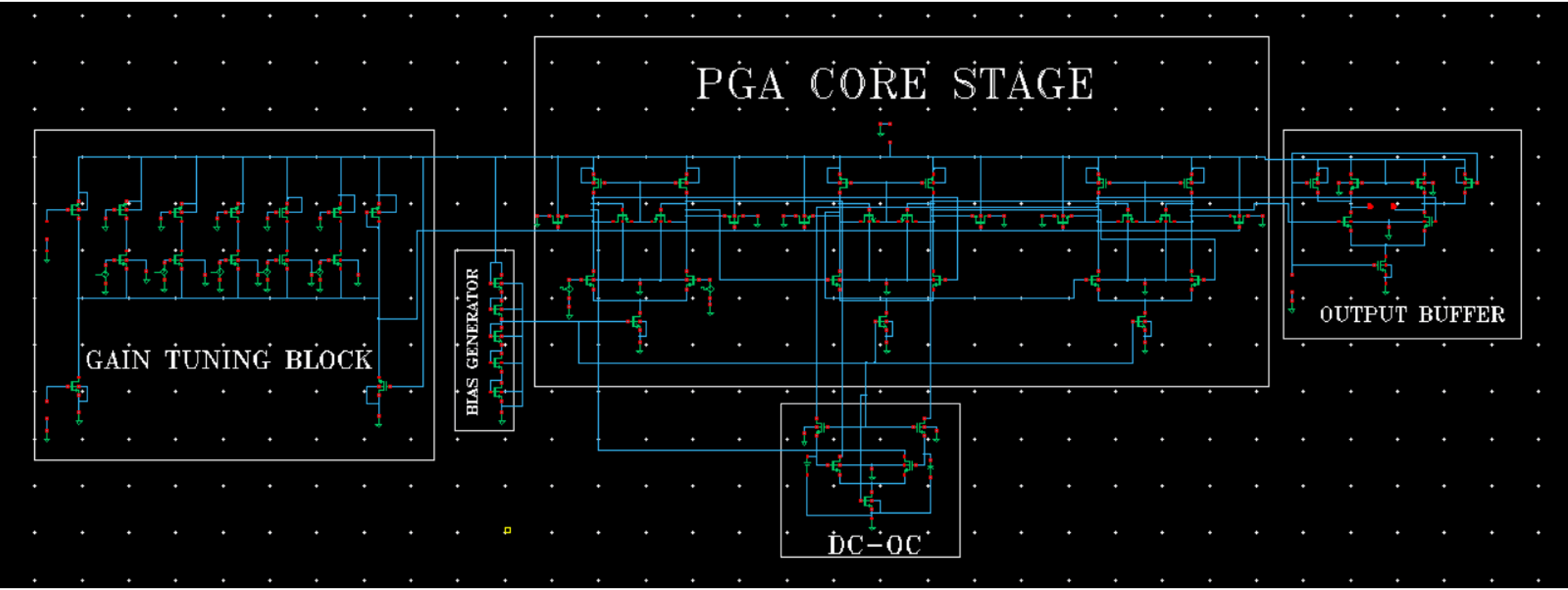
Vin_dc=562mv

Vtune=150mv−450mv

V_bias=590mv



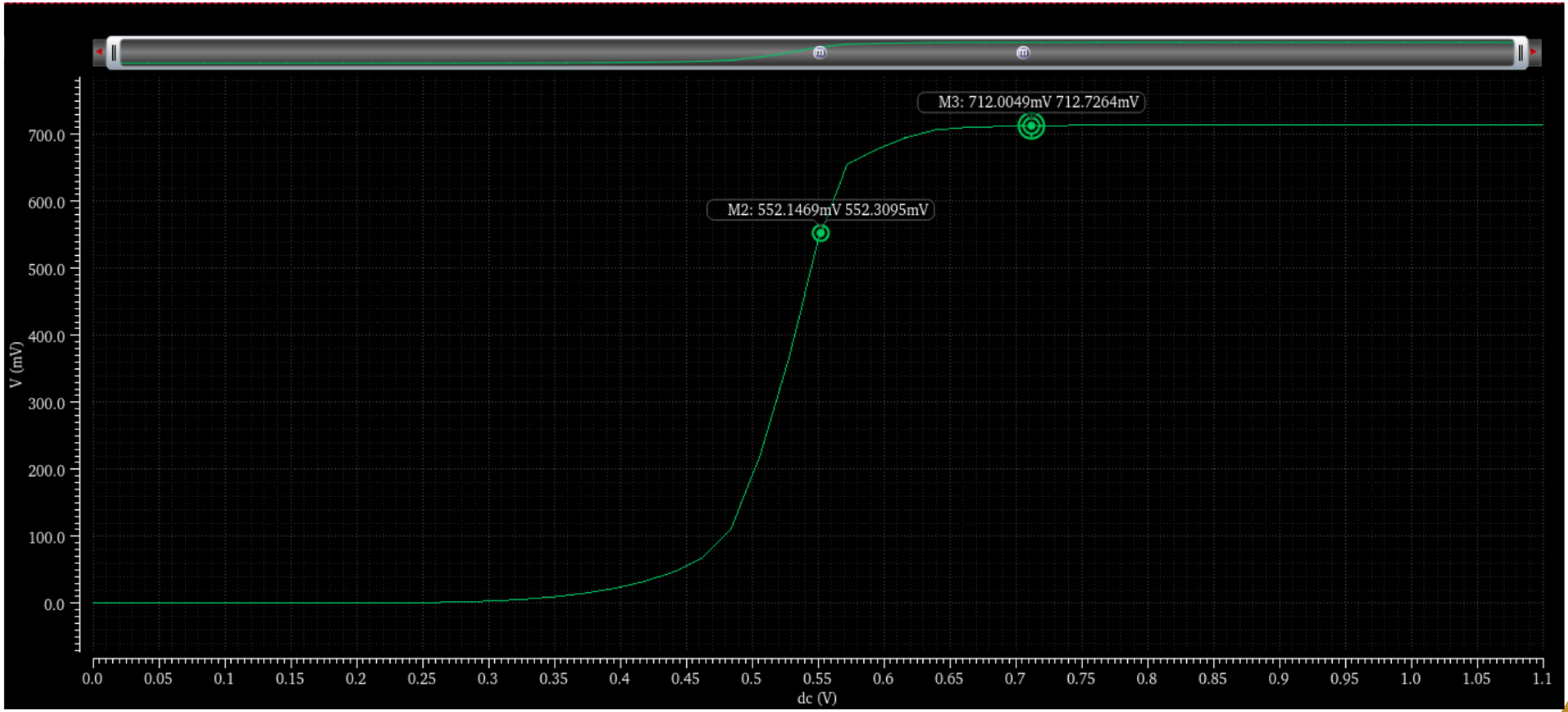




PGA CORE STAGE
GAIN TUNING BLOCK
BIAS GENERATOR
OUTPUT BUFFER
DC-OC

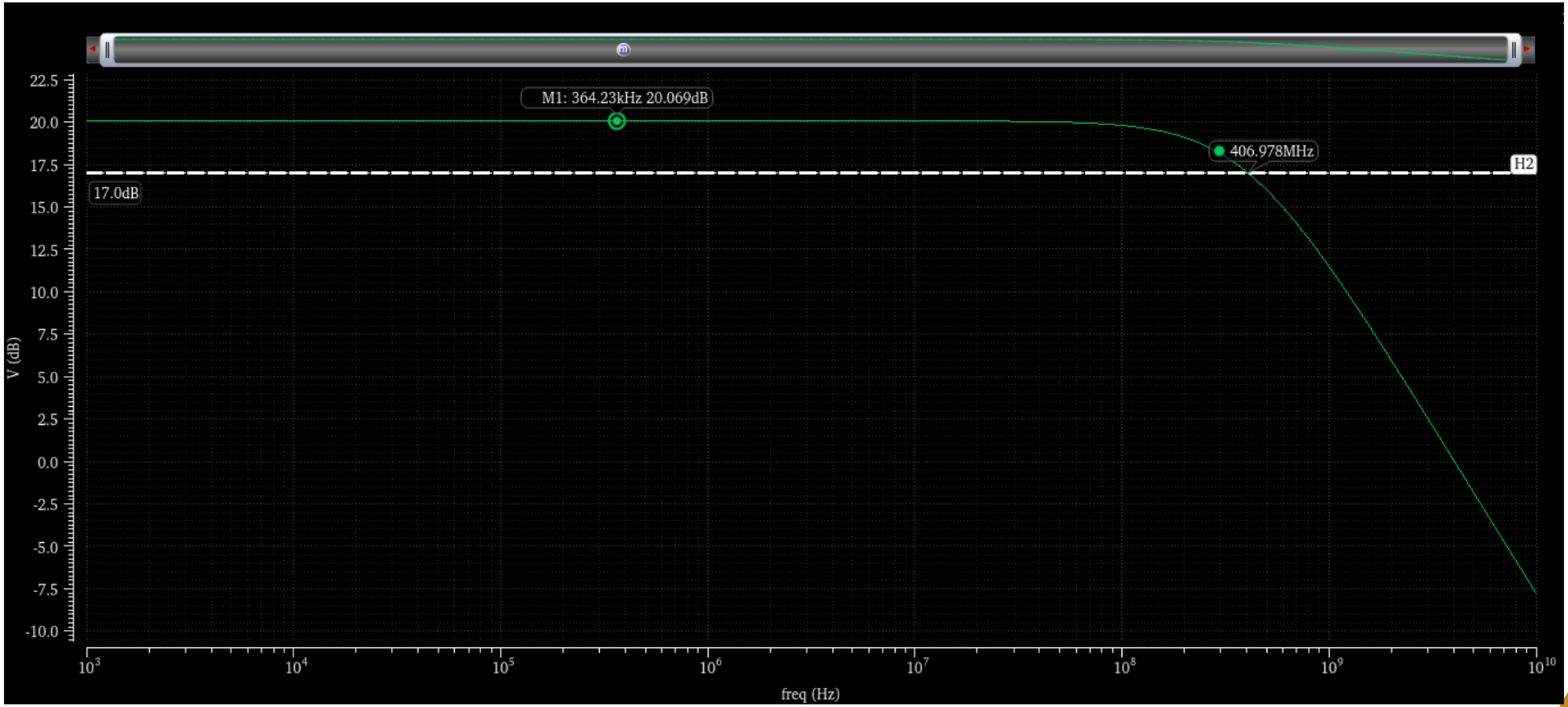
Observation and Results

Transfer Characteristics of Single cell PGA :



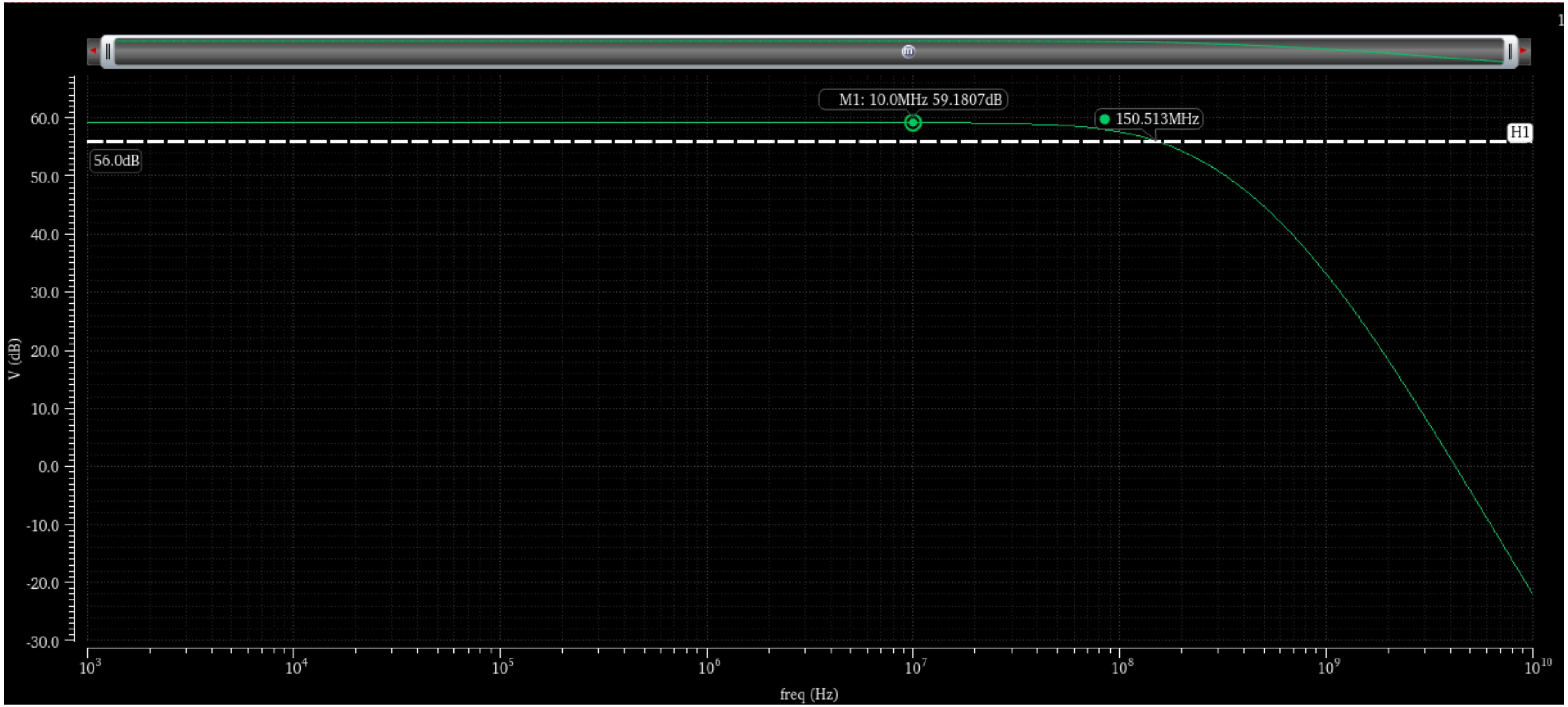
Observation and Results

Bode Plot of single PGA :



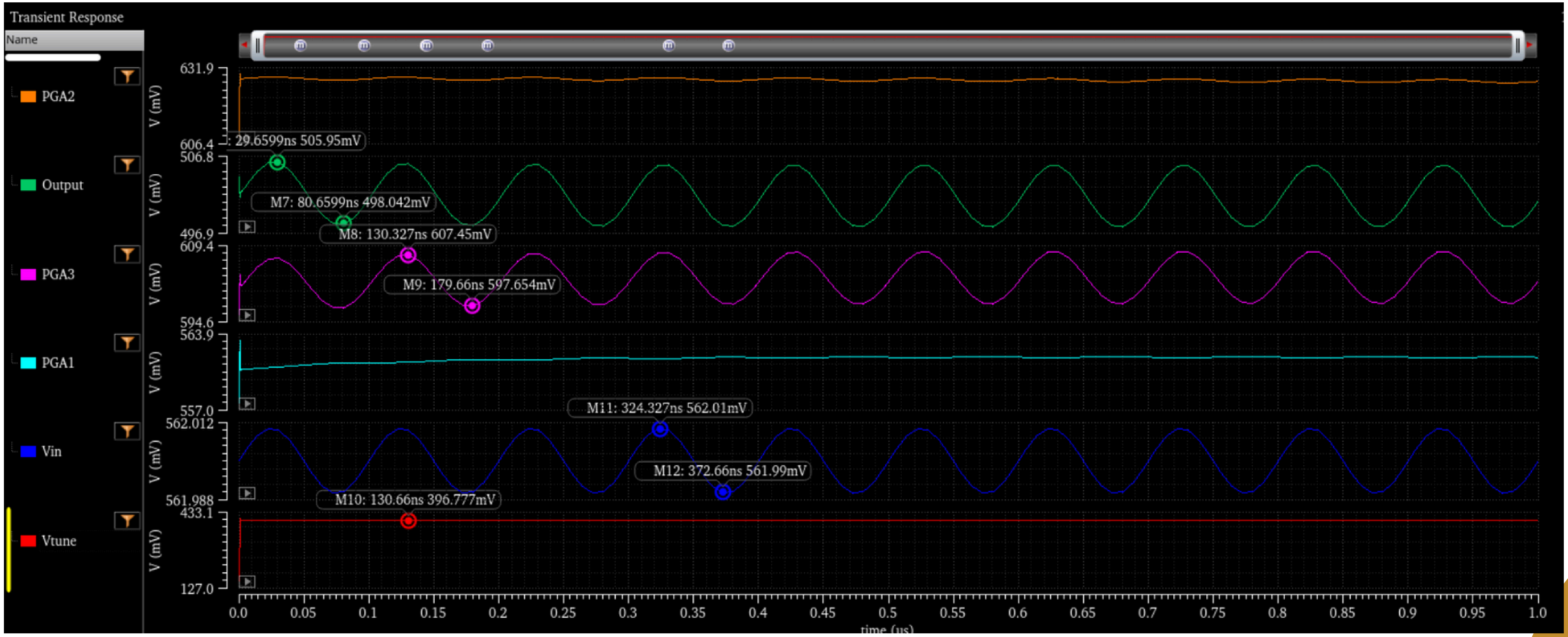
Observation and Results

3 Stage PGA Bode Plot :



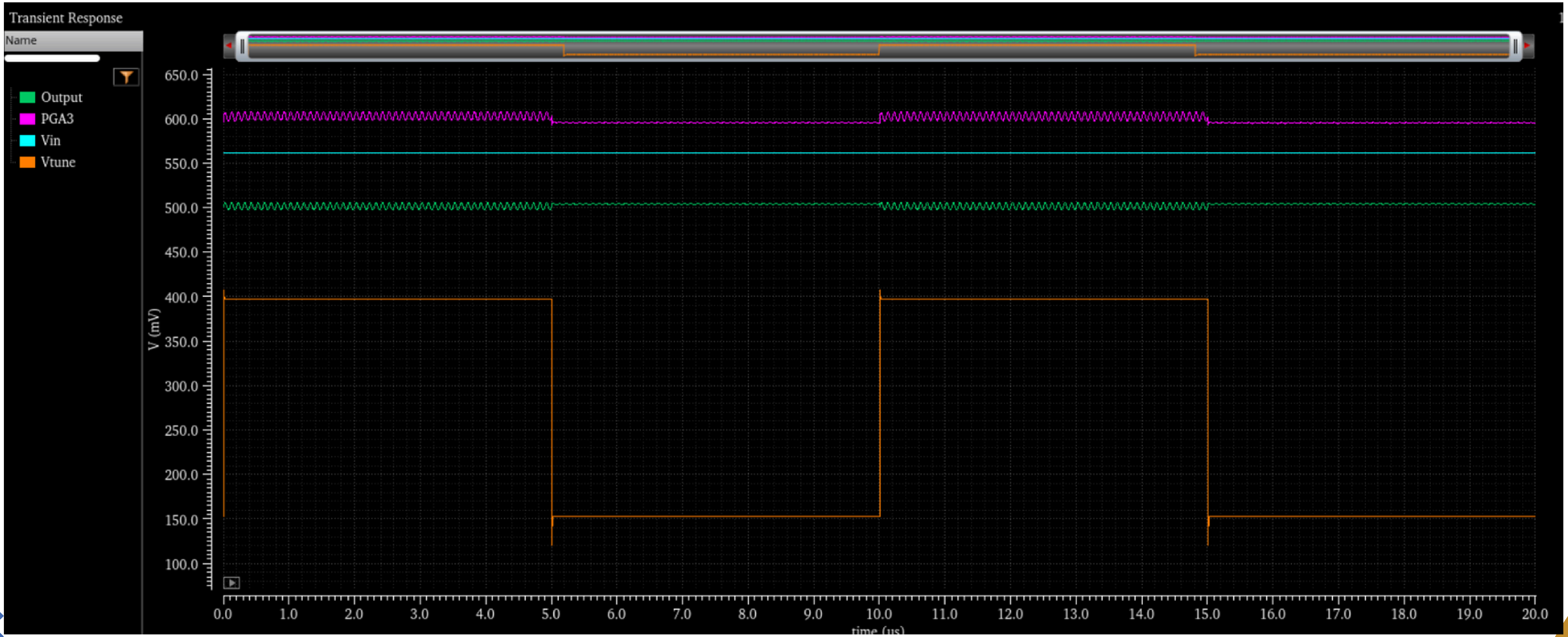
Observation and Results

Output waveforms of Binary (b’11111):



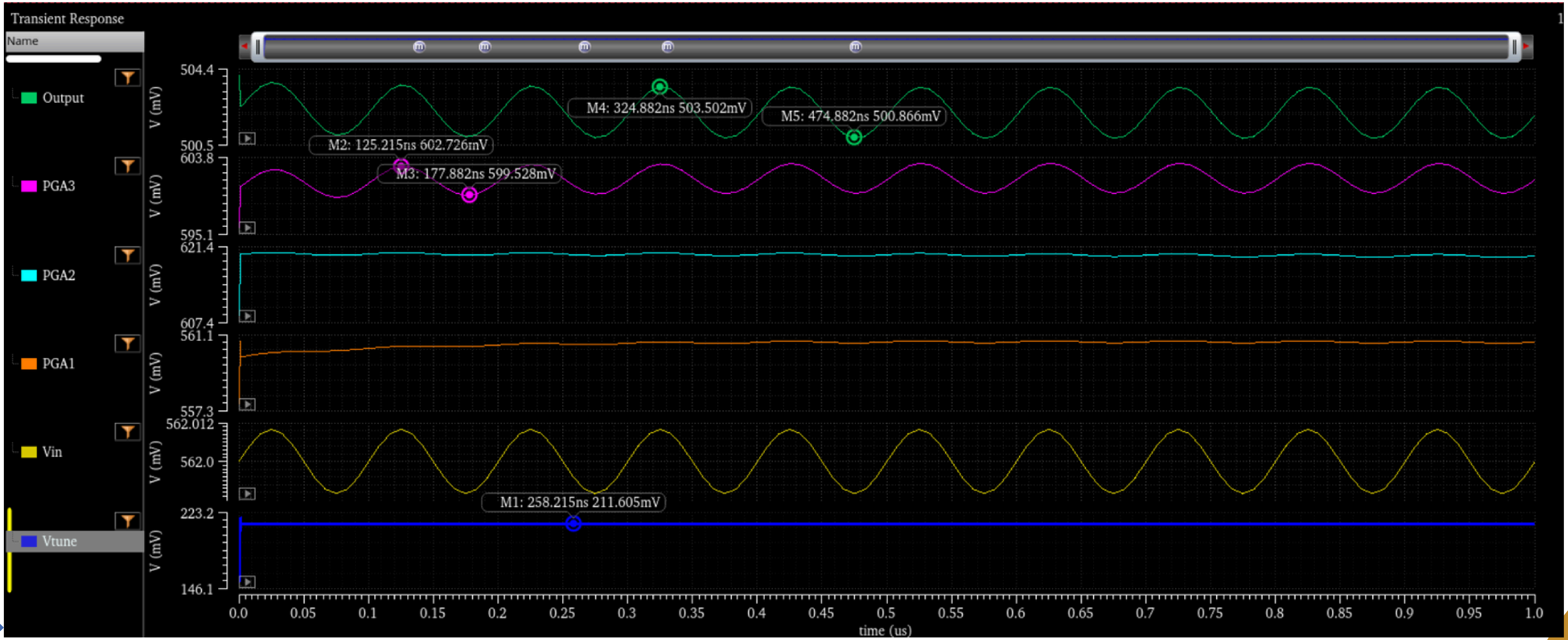
Observation and Results

Output waveforms of Binary (b’11111):



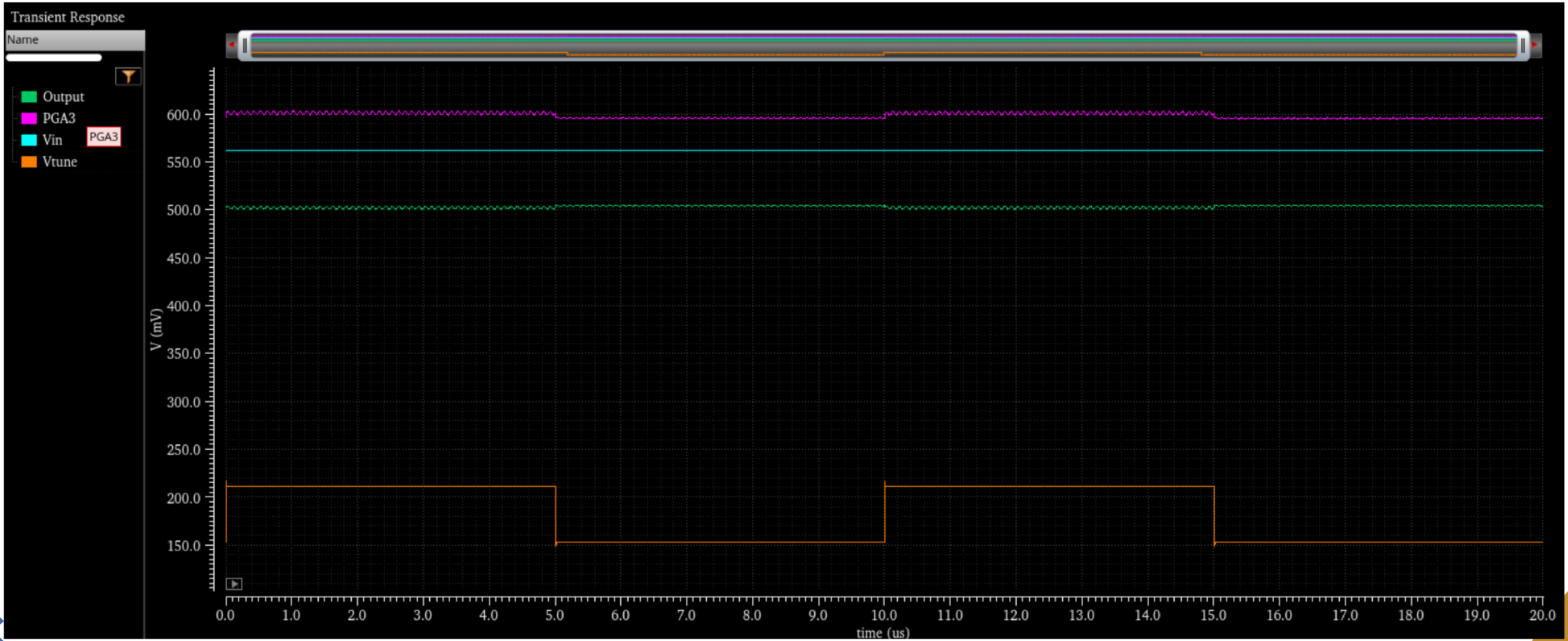
Observation and Results

Output waveforms of Binary (b'00100):



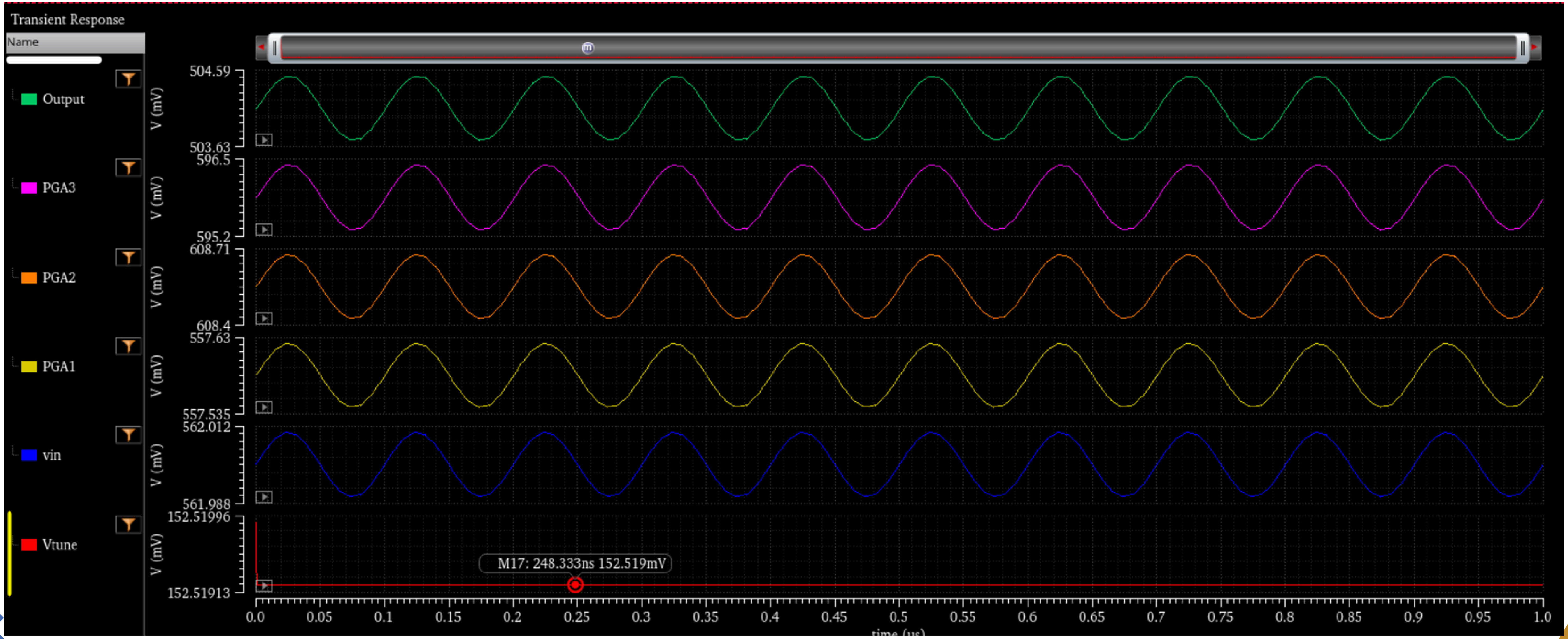
Observation and Results

Output waveforms of Binary (b’00100):



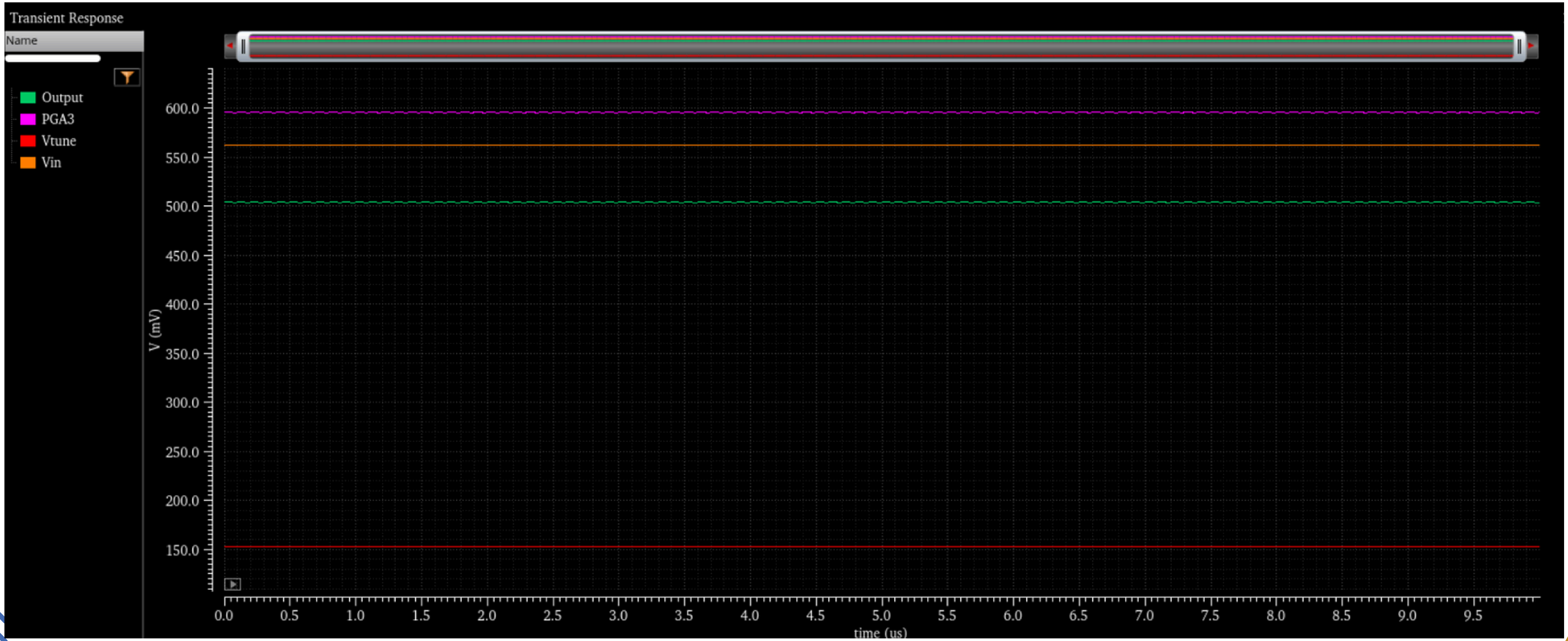
Observation and Results

Output waveforms of Binary (b’00000):



Observation and Results

Output waveforms of Binary (b’00000):



Observation and Results

Bode plot of 3 stage complete PGA circuit